Complete Experiment Log

XLE Energy ETF Direction Model

All Sessions — 74 Experiments

energy-autoresearch

May 2026

Dataset: XLE (Yahoo Finance, 2020-01-01 onward), 80/20 chronological split. **Metric:** Annualised Sharpe $= \sqrt{252} \cdot \mathbb{E}[\text{sign}(\hat{s}) \cdot r_t] / \text{std}(\cdot)$. WF Sharpe is mean walk-forward Sharpe over expanding folds ("—" = not computed). Status: **best** = best committed model; keep = improved over prior; — = discarded. Sessions: Det-S1 = alternative-data experiments (Apr 20); S1–S3 = autoresearch loop; Sweep = HP grid; Post = post-cleanup; Ablation = structured Phase 2 ablations.

Table 1: Complete experiment log. TW = train window (trading days); WTI = shock filter threshold; Val S = validation Sharpe; WF S = walk-forward Sharpe.

Run ID	Sess.	Date	Description	Model	n	d	msl	TW	WTI	Val S	WF S	Status
Det-S1												
det_001	Det-S1	2026-04-20	OLS on 24 alternative-data features	OLS (statsmode	—	—	—	full	—	-0.7390	—	ref
det_002	Det-S1	2026-04-20	OLS selected+engineered (10 features, trend/lag)	OLS (selected)	—	—	—	full	—	0.0390	—	—
det_003	Det-S1	2026-04-20	Ridge / Lasso with cross-validated alpha	Ridge/Lass	—	—	—	full	—	0.6130	—	keep
det_004	Det-S1	2026-04-20	RandomForest n=300, d=4, msl=15 (regressor)	RF Regressor	300	4	15	full	—	-0.7050	—	—
det_006	Det-S1	2026-04-20	Lasso + signal threshold sweep; 5-fold walk-forward	Lasso+thre	—	—	—	full	—	0.6130	-0.2405	keep
det_007	Det-S1	2026-04-20	Signal audit: 15 non-financial signals individually	Individual signals	—	—	—	full	—	1.6870	—	keep
det_008	Det-S1	2026-04-20	Normalised signal composite (EW + OLS-weighted)	Signal composite	—	—	—	full	—	0.6630	—	keep
det_009	Det-S1	2026-04-20	Lasso fine alpha grid (40-pt, 1e-5 to 0.1) on top signals	Lasso (fine grid)	—	—	—	full	—	0.6130	—	keep
det_010	Det-S1	2026-04-20	Walk-forward: best signal + Lasso ensemble (5-fold)	Signal+Las WF	—	—	—	full	—	1.0420	0.5733	keep
S1												
tsv_002	S1	2026-04-16	Pre-session smoke test	RandomFore	—	—	—	756	—	0.8347	—	smoke
tsv_003	S1	2026-04-16	Pre-session smoke test	RandomFore	—	—	—	756	—	0.8347	—	smoke
tsv_004	S1	2026-04-18	LinearRegression baseline	LinearRegr	—	—	—	756	—	0.8584	—	ref
tsv_005	S1	2026-04-18	RF: RetFeatures + Ridge	Ridge	—	—	—	756	—	0.0653	—	—
tsv_006	S1	2026-04-18	GradientBoostingRegressor	GradientBo	—	—	—	756	—	0.1524	—	—
tsv_007	S1	2026-04-18	Rolling z/RSI/vol + Ridge	Ridge	—	—	—	756	—	0.4703	—	—
tsv_008	S1	2026-04-18	Rolling feat + LogisticReg	LogisticRe	—	—	—	756	—	0.9581	—	keep
tsv_009	S1	2026-04-18	LogReg + lag2/lag3/MACD/RSI	RandomFore	—	—	—	756	—	0.8824	—	—

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Run ID	Sess.	Date	Description	Model	n	d	msl	TW	WTI	Val S	WF S	Status
tsv_010	S1	2026-04-18	RF depth=3, msl=30	RandomFore	—	3	30	756	—	1.2623	—	keep
tsv_011	S1	2026-04-18	RF d=2, msl=25, sqrt, n=200	RandomFore	200	2	25	756	—	1.3092	—	keep
tsv_012	S1	2026-04-18	MLPClassifier (128,) alpha=1	MLPClassif	—	—	—	756	—	1.1365	—	—
tsv_013	S1	2026-04-18	RF n=300, d=2, msl=20	RandomFore	300	2	20	756	—	1.3717	—	keep
tsv_014	S1	2026-04-18	RF n=200, d=2, msl=22	RandomFore	200	2	22	756	—	1.3954	—	**best-S1**
tsv_015	S1	2026-04-18	Direct Sharpe optimizer (tanh)	SharpeOpti	—	—	—	756	—	0.0267	—	—
tsv_016	S1	2026-04-18	Restored RF n=200, d=2, msl=22	RandomFore	200	2	22	756	—	1.3954	—	keep
tsv_001	S1	ÿ2026-04-	Pre-session smoke test	RandomFore	—	—	—	756	—	0.8220	—	smoke
S2												
tsv_017	S2	2026-04-25	Session 2 start	RandomFore	—	—	—	756	—	1.1685	—	smoke
tsv_018	S2	2026-04-25	S2 intermediate	RandomFore	—	—	—	756	—	0.7420	—	—
tsv_019	S2	2026-04-25	S2 intermediate	RandomFore	—	—	—	756	—	0.7372	—	—
tsv_020	S2	2026-04-25	S2 intermediate (+roll window)	RandomFore	—	—	—	756	—	1.8278	—	keep
tsv_021	S2	2026-04-25	WTI shock filter thresh=3%	RandomFore	—	—	—	756	3%	2.0219	—	keep
tsv_022	S2	2026-04-25	WTI shock filter thresh=2%	RandomFore	—	—	—	756	2%	2.2119	—	**best**
tsv_023	S2	2026-04-25	WTI shock filter thresh=1.5%	RandomFore	—	—	—	756	2%	2.0065	—	—
tsv_024	S2	2026-04-25	WTI ret1+mom5 as RF inputs	RandomFore	—	—	—	756	—	1.3403	—	—
tsv_025	S2	2026-04-25	Contrarian signal on shock days	SignalComp	—	—	—	756	—	1.9532	—	—
Sweep												
run_001	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=22, win=20, wti=0.02, tw=756	RandomFore	300	2	22	756	2%	2.5903	—	keep
run_002	Sweep	2026-04-25	HP sweep: n=300, d=3, msl=22, win=20, wti=0.02, tw=756	RandomFore	300	3	22	756	2%	2.5941	—	keep
run_003	Sweep	2026-04-25	HP sweep: n=300, d=4, msl=22, win=20, wti=0.02, tw=756	RandomFore	300	4	22	756	2%	2.0454	—	—
run_004	Sweep	2026-04-25	HP sweep: n=300, d=1, msl=22, win=20, wti=0.02, tw=756	RandomFore	300	1	22	756	2%	0.7120	—	—
run_005	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=10, win=20, wti=0.02, tw=756	RandomFore	300	2	10	756	2%	2.6104	—	keep
run_006	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=50, win=20, wti=0.02, tw=756	RandomFore	300	2	50	756	2%	1.7419	—	—
run_007	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=100, win=20, wti=0.02, tw=756	RandomFore	300	2	100	756	2%	1.8515	—	—
run_008	Sweep	2026-04-25	HP sweep: n=100, d=2, msl=22, win=20, wti=0.02, tw=756	RandomFore	100	2	22	756	2%	2.5363	—	—
run_009	Sweep	2026-04-25	HP sweep: n=600, d=2, msl=22, win=20, wti=0.02, tw=756	RandomFore	600	2	22	756	2%	2.5990	—	keep
run_010	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=22, win=10, wti=0.02, tw=756	RandomFore	300	2	22	756	2%	2.5903	—	—
run_011	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=22, win=40, wti=0.02, tw=756	RandomFore	300	2	22	756	2%	2.5903	—	—

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Run ID	Sess.	Date	Description	Model	n	d	msl	TW	WTI	Val S	WF S	Status
run_012	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=22, win=20, wti=0.03, tw=756	RandomFore	300	2	22	756	3%	2.3902	—	—
run_013	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=22, win=20, wti=0.01, tw=756	RandomFore	300	2	22	756	1%	1.5038	—	—
run_014	Sweep	2026-04-25	HP sweep: n=300, d=2, msl=22, win=20, wti=0.02, tw=504	RandomFore	300	2	22	504	2%	1.2833	—	—
run_015	Sweep	2026-04-25	HP sweep: n=300, d=3, msl=15, win=10, wti=0.02, tw=756	RandomFore	300	3	15	756	2%	2.4010	—	—
S3												
tsv_026	S3	2026-04-30	S3 early probe	RandomFore	—	—	—	756	—	0.7453	—	smoke
tsv_027	S3	2026-05-06	S3 early probe	RandomFore	—	—	—	756	—	1.6544	—	smoke
tsv_028	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.0925	—	keep
tsv_029	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.9338	—	—
tsv_030	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.6980	—	—
tsv_031	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.3099	—	—
tsv_032	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	0.7485	—	—
tsv_033	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.9362	—	keep
tsv_034	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.7698	—	—
tsv_035	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.6660	—	—
tsv_036	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.5416	—	—
tsv_037	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.7753	—	keep
tsv_038	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.1332	—	—
tsv_039	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.0365	—	keep
tsv_040	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.1316	—	keep
tsv_041	S3	2026-05-07	S3 peak: msl tune	RandomFore	—	—	—	756	—	2.1982	—	**best-S3**
tsv_042	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.9865	—	—
tsv_043	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.0790	—	keep
tsv_044	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	1.0358	—	—
tsv_045	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.1776	—	keep
tsv_046	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.1563	—	keep
tsv_047	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.0418	—	keep
tsv_048	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	2.0526	—	keep
tsv_049	S3	2026-05-07	S3 run	RandomFore	—	—	—	756	—	0.4453	—	—
Post												

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Run ID	Sess.	Date	Description	Model	n	d	msl	TW	WTI	Val S	WF S	Status
tsv_050	Post	2026-05-18	Post-cleanup smoke test	RandomFore	—	—	—	756	—	2.5903	—	smoke
tsv_051	Post	2026-05-18	XGBoost baseline	XGBoost	—	—	—	756	—	0.5408	-0.3981	—
tsv_052	Post	2026-05-18	XGBoost d=2, n=800	XGBoost	800	2	—	756	—	0.2908	-0.6807	—
tsv_053	Post	2026-05-18	RF baseline + walk-forward	RandomFore	—	—	—	756	—	2.5949	0.3226	keep
Ablation												
exp_001	Ablatior	2026-05-27	Baseline: LinearRegression on ret_lag only	LinearRegr	—	—	—	—	—	0.5932	0.0000	ablation
exp_002	Ablatior	2026-05-27	RF direction classifier — Session 1 best (n=200, depth=2, msl=22, no shock filter)	RandomFore	200	2	22	756	9900%	1.8282	0.3370	ablation
exp_003	Ablatior	2026-05-27	RF sweep baseline (n=300, depth=2, msl=22, wti_thresh=2%)	RandomFore	300	2	22	756	2%	1.9616	0.3012	ablation
exp_004	Ablatior	2026-05-27	RF depth ablation (n=300, depth=4, msl=22, wti_thresh=2%) — run_003	RandomFore	300	4	22	756	2%	2.4791	0.1485	ablation
exp_005	Ablatior	2026-05-27	RF best sweep (n=600, depth=2, msl=22, wti_thresh=2%) — run_009	RandomFore	600	2	22	756	2%	2.2959	0.1998	ablation
exp_006	Ablatior	2026-05-27	RF shock-filter ablation — no WTI filter (n=300, depth=2, msl=22)	RandomFore	300	2	22	756	9900%	1.8296	0.4611	ablation
exp_007	Ablatior	2026-05-27	RF training-window ablation (n=300, depth=2, msl=22, train_window=504, wti_thresh=2%)	RandomFore	300	2	22	504	2%	1.0928	0.2460	ablation